



Do teams need both hands? An analysis of team process ambidexterity and the enabling role of information technology

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ABSTRACT

As a core organizational resource, business processes are vital for organizational teams. To deal with today's volatile business environment, organizations need to be ambidextrous in terms of process capabilities. However, little is known about how process ambidexterity, process standardization, and process agility, are enabled by information technology (IT) and related to team-level activities. To fill this gap in the literature, we conducted a field study based on 160 teams of 1081 individuals from seven companies in South Korea. Our results show that IT enables both process standardization and agility, and that a team's process ambidexterity has a positive effect on inter-team coordination and team innovation, which in turn have a direct impact on team performance. Our findings highlight the importance of process ambidexterity by investigating the enabling role of IT and its outcomes in a team. Our results offer theoretical and practical implications from the perspective of team process ambidexterity.

1. Introduction

Business processes influence every behavior of organizational actors, including teams. By designing and executing processes, organizations accomplish their desired purposes and objectives (Ray, Barney, & Muhanna, 2004). As business processes have become valuable assets leading to improvements in organizational performance, organizations have introduced a variety of methods for business process management (BPM), such as process standardization, which is a proven long-term process management method. Standardized processes enable companies to align organizational resources with organizational units to improve overall operational efficiency and accuracy (Lee, DeLone, & Espinosa, 2010). Furthermore, companies can benefit from process standardization by using well-defined templates to guarantee transparent and uniform process execution (Wüllenweber, Beimborn, Weitzel, & König, 2008); minimize variances and uncertainties (Tregear, 2015); and optimize time, cost, and quality (Münstermann, Eckhardt, & Weitzel, 2010).

Although well-established standards can certainly improve the operational efficiency of business processes, adapting to the ever-changing business environment has become a critical capability for organizations to survive; thus, business processes need to be adaptable to changes (Chen et al., 2014; Shukla, 2020). This capability is called process

agility and refers to the ability to adjust business processes in response to changes (Tallon, 2008). The importance of process agility has been growing, as proactive response to market changes has strong relevance to business performance (Raschke, 2010) and how firms deal with the market environment is critical to their survival (Van Oosterhout, Waarts, & van Hillebergersberg, 2006).

However, having only one good quality is not sufficient for creating organizational values for survival. Managers and researchers have pointed out that pursuing two different, or ambidextrous, qualities is important for business performance despite being a challenging task (He & Wong, 2004). With respect to process management, managing the ambidexterity of process capabilities can be a critical task, as business processes should be able to both enhance the efficiency of execution and adapt to changes in the external environment (Lee et al., 2010). Even though having different process capabilities—for instance, process standardization and process agility—simultaneously is difficult, both are essential for team operations. Although the importance of process ambidexterity has grown (Rosemann, 2014), the issue has not received sufficient empirical attention.

Information technology (IT) is worth noting for enabling business processes, as it plays an important role in organizational processes with respect to transformation of resources into capabilities, which is critical to creating an organization's competitive advantages (Kohli & Grover,

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